MILITARY SPECIFICATION

PALLET, METAL, FOR METAL CONTAINER FOR 105MM AND 120MM TANK AMMUNITION

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabrication of a number of the selected items, quality assurance provisions and packaging of the special metal pallets for tank ammunition pallets. The special pallets will consist of a base and a cover that will unitize the selected quantity and type of the metal container for the 105MM and 120MM tank ammunition cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 8140
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3.7.2 Paint coating. The paint coating shall be in accordance with applicable drawings and comply with instructions in MIL-STD-171. The finish coat shall cover all required surfaces and be continuous with the possible exception of a few light scratches not exposing the base metal or the primer coat.

3.7.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection . Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractors's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article sample shall consist of dash numbered assemblies as per dwg AC20000353 listed in Table I, of this specification.

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MILITARY

MIL-W-12332	-	Welding Resistance, Spot, Seam and Production, for fabricating Assemblies of Low-Carbon Steel
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1261	-	Welding Procedures for Constructional Steel

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

US Army Defense Ammunition Center and School

AC20000353 - Pallet, Metal, for Metal Containers

(Copies of specifications, standards and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E8	-	Tension Testing of Metallic Materials, Standard Method of
ASTM-D3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. However, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Pallets. The pallet assemblies shall comply with all requirements specified on drawing (dwg) AC20000353 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Stacking load. The pallets shall be able to withstand a 10,000 pound stacking load.

3.4 Welding. The welding shall comply with the requirements of MIL-W-12332 or MIL-STD-1261.

3.4.1 Tensile strength. The tensile strength of the weld shall equal or exceed the tensile strength of the material.

3.5 Painted surface. The painted surfaces of the pallets shall comply with the requirements of TT-C-490.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Pallets. The pallets shall be regular, smooth, and free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability, safety and appearance of the pallets.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractors's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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b. Quality Conformance Inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article sample shall consist of dash numbered assemblies as per dwg AC20000353 listed in Table I, of this specification.

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TABLE I. First article quantities

DRAWING DASH NO.	PALLET CONFIGURATION OF METAL CONTAINERS	QUANTITIES REQUIRED OF-FIRST ARTICLE	METAL CONTAINER NOMENCLATURE
-1	6 Wide by 5 High	8	105MM Light Weight Container
-2	6 Wide by 5 High	8	105MM PA 104 Container
-3	5 Wide by 6 High	6	120MM Light Weight Container

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48663 (AR)

PARAGRAPH	TITLE: Metal Pallet		SHEET 1 OF 1		DRAWING NUMBER: AC20000353
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Pallet				
	a. Examination for Defects	*		3.2	4.4.2.1
	b. Stacking strength	*		3.3	4.5.2

NOTES: *As listed in Table I of this specification

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. 100% inspection may be performed at the discretion of the Contracting Officer's Representative (COR).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48663 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Metal Pallet				AC20000353
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Metal thickness		1.0%	3.2	Gage
103	Pallet weight		1.0%	3.2	Scale
104	Overall length		1.0%	3.2	Gage
105	Overall width		1.0%	3.2	Gage
106	Overall height		1.0%	3.2	Gage
107	Side rail slots alignment		1.0%	3.2	Gage
108	Container stacking lug hole alignment		1.0%	3.2	Gage
109	Weld tensile	(b)		3.4.1	4.5.1
110	Primer thickness		1.0%	3.2	Gage
111	Paint thickness		1.0%	3.2	Gage
112	Marking missing, incomplete or illegible		1.0%	3.2	Visual
Minor					
201	Protective finish damaged, bare spots, exposed metal, etc.		1.0%	3.2	Visual
202	Poor workmanship		1.0%	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing.

4.4.3.1 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.3.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM-E8. Three specimens of each gage thickness shall be selected from the material used to fabricate the pallets for determination of the material tensile strength. Also, the tensile strength of the welds will be determined by applying the force at right angles to the weld. Each of these values shall be compared with the lowest determined value of the body material tensile strength for compliance with the requirement of 3.4.1. If the samples cannot meet these requirements the lot will be rejected.

4.5.2 Stacking strength. The number of samples, as listed in Table I of the specification, shall be randomly selected for this test. The pallet shall be loaded, to simulate a stack of identical pallet loads, to the stacking load requirement for a period of 1 hour minimum. If any pallet fails to meet the requirement, the lot shall be rejected.

5. PACKAGING

5.1 Preservation and packaging. None required.

5.2 Packing.

5.2.1 Commercial. The metal pallet shall be packed in accordance with ASTM-D-3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the pallets. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If pallets are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

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5.4 Shipping. When pallets from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

6.1 Intended use. This metal pallet is intended for use with the metal tank ammunition container for the 105MM and 120MM tank ammunition cartridge.

6.2 Ordering data. See MIL-A-48078

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I(D), Dover, New Jersey, 07801-5001.

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-A(D), Dover, New Jersey 07801-5001. This description should include but not be limited to the accuracy and precision and drawings of any special equipment required.

6.5 Subject term (key word) listing.
Ammunition
Cannon
Cartridge
Container
Metal
Pallet
Tank

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A718)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48663	PALLET, METAL, FOR METAL CONTAINER FOR 105MM, and 120MM TANK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
AMMUNITION	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

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